

# PROJECT REPORT

TEAM ID:

TITLE: Comprehensive Analysis and Dietary Strategies with Tableau:

A College Food Choices Case Study

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## 1. INTRODUCTION

### 1.1 Project Overview

As a college health and wellness administrator, my mission is to proactively support the nutritional well-being of students. However, our current understanding of student dietary behaviours is hindered by fragmented, static, or siloed data. Without timely, actionable insights, our ability to address nutritional deficiencies and trends in real-time is limited - leaving us in a largely reactive position.

### 1.2 Purpose

The primary purpose of this project is to develop an interactive, data-driven platform using Tableau that enables college wellness administrators to:

- 1 Understand and track student eating behaviours in real-time.
- 2 Identify nutritional deficiencies and food trends across the student population.
- 3 Enhance collaboration between health services and dining operations.
- 4 Support evidence-based wellness programs and menu improvements that promote healthier eating on campus.

## 2. IDEATION PHASE

2.1 Problem Statement I am a college health and wellness administrator I'm trying to understand and improve the dietary and nutritional habits of students But I lack real-time, visual insights into eating behaviors, health trends, and deficiencies Because the

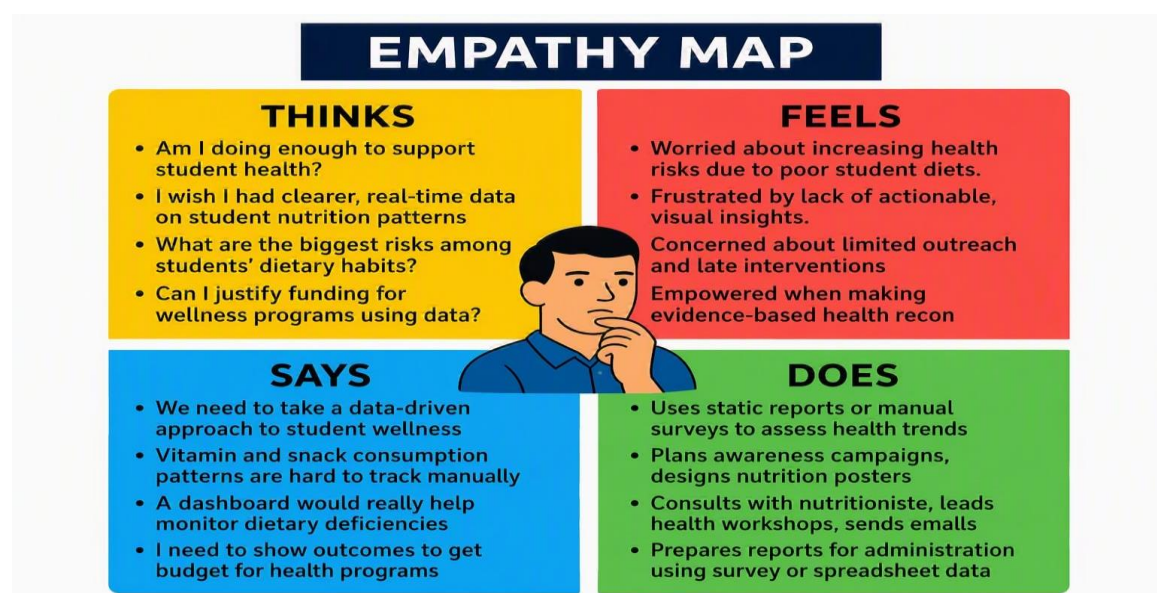
data is scattered, static, or underutilized for strategic wellness interventions Which makes me feel reactive and under-equipped to support student health proactively.

PS-1: I am a student affairs officer who wants to monitor student nutrition habits to prevent health issues, but I don't have access to interactive tools that give real-time data on students' dietary trends.

PS-2: I am a university nutritionist trying to detect deficiencies in students' diets, but it's difficult to pinpoint problem areas without an integrated platform that shows snack overconsumption, vitamin neglect, and health perception metrics.

PS-3: I am a wellness strategist aiming to develop personalized health plans for students, but predictive analytics tools are not in place to proactively identify high- risk individuals and suggest tailored interventions.

## 2.2 Empathy Map



## 2.3 Brainstorming

| Grouping Category          | Ideas Generated                                                      |
|----------------------------|----------------------------------------------------------------------|
| Nutritional Monitoring     | Track fruit and vegetable intake by gender, age, and exercise levels |
| Deficiency Detection       | Find gaps in vitamin intake and highlight high snack consumption     |
| Predictive Health Insights | Categorize students into risk groups (High, Moderate, Low)           |
| Visualization Tools        | Use bar charts, scatter plots, heatmaps, and KPI cards in Tableau    |
| Real-Time Alerts           | Send alerts for low fruit/veggie intake or high snack calories       |

|                            |                                                                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personalized Plans</b>  | Suggest meal plans and health tips based on risk levels                                                                                                                           |
| <b>Awareness Campaigns</b> | Create educational programs using insights from dashboards                                                                                                                        |
| <b>Priority Level</b>      | <b>Ideas</b>                                                                                                                                                                      |
| <b>High</b>                | <ul style="list-style-type: none"> <li>✓ Build Tableau dashboards for real-time intake monitoring</li> <li>✓ Classify students by risk using fruit and exercise data .</li> </ul> |
| <b>Medium</b>              | <ul style="list-style-type: none"> <li>✓ Visualize snack consumption vs. health perception</li> <li>✓ Track vitamin intake by demographic .</li> </ul>                            |
| <b>Low</b>                 | <ul style="list-style-type: none"> <li>✓ Connect Tableau to live cafeteria menu data (future use)</li> <li>✓ Add AI-based meal suggestions (after MVP stage) .</li> </ul>         |

### 3.REQUIREMENT ANALYSIS

#### 3.1 Customer Journey Map

# CUSTOMER JOURNEY MAP

| JOURNEY STAGE                                                                                        | AWARENESS                                                             | CONSIDERATION                                                             | DECISION                                                  | ACTION                                                                      | POST-ACTION                                                   |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------|
| <br>Customer Goal   | Explore tools to visualize and track student nutrition patterns       | Choose a data visualization platform to monitor in improve dietary trends | Implement visual dashboards and monitor student data      | Evaluate impact adjust strategies, and plan next interventions              | Assess results and refine strategies for future programs      |
| <br>Touchpoints     | Research on tools (eg. Tableau, workshops, health committee meetings) | Tableau trials, case studies, peer recommendations                        | Tableau dashboards, KPI alerts, student reports           | Follow-up dashboards, wellness program feedback                             | Ongoing dashboards, wellness program feedback                 |
| <br>Experience      | Curious but overwhelmed by data complexity                            | Confident in Tableau's interactive dashboard capabilities                 | Empowered to make decisions based on trends and alerts    | Reflects on dashboard insights and makes recommendations for next semester  | Satisfied seeing measurable improvements and long-term impact |
| <br>Pain Points    | No integrated system for diet tracking                                | Budget limitations, training needs for Tableau                            | Time consuming data cleaning, interpreting visualizations | Difficulty measuring long-term change                                       | Maintaining engagement and continuous data quality            |
| <br>Opportunities | Educate stakeholders about dietary data value                         | Show past success stories and predicted impact                            | Automate data updates and schedule alerts                 | Expand dashboards with advanced analytics and student:level personalization | Enhance reporting and share insights with more stakeholders   |

## 3.2 Solution Requirements

### Functional requirements:

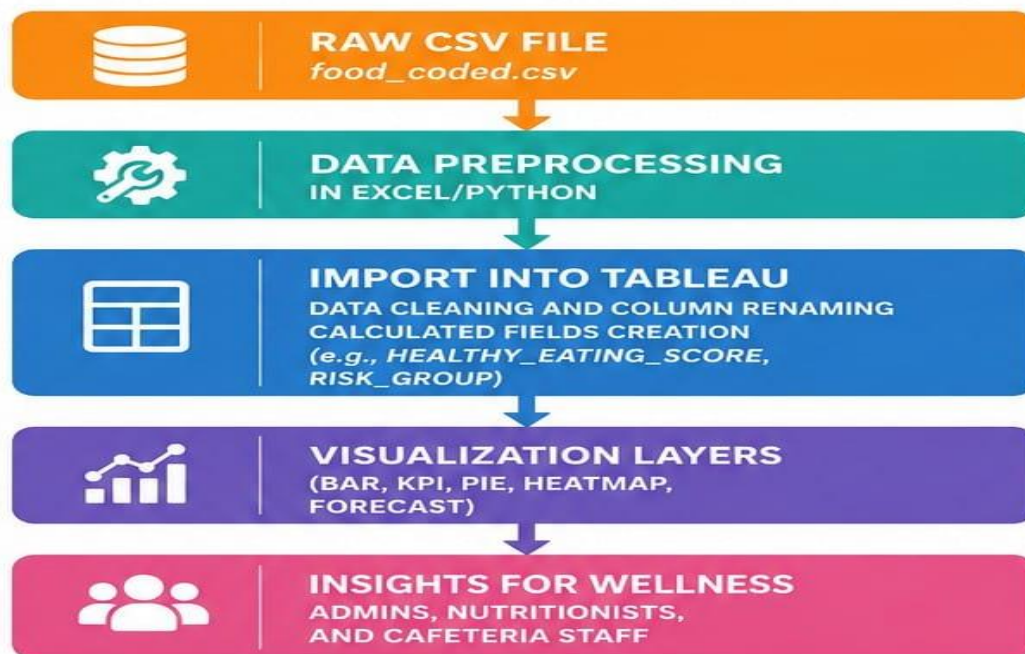
| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task) |
|--------|-------------------------------|------------------------------------|
| FR-1   | User Registration             | Registration through Form          |
|        |                               | Registration through Gmail         |
|        |                               | Registration through LinkedIn      |
| FR-2   | User Confirmation             | Confirmation via Email             |
|        |                               | Confirmation via OTP               |
| FR-3   | Data Visualization            | Generate Text Table                |
|        |                               | Generate Horizontal Bar Chart      |
|        |                               | Generate Side-by-Side Bar          |
|        |                               | Generate Tree Map                  |

|             |                            |                                     |
|-------------|----------------------------|-------------------------------------|
|             |                            | Generate Dual Line Chart            |
|             |                            | Generate Packed Bubble Sort         |
| <b>FR-4</b> | Data Filtering and Sorting | Filter by Data Range, Measure Names |
|             |                            | Sort by Calculations                |
|             |                            | Filter by Gender                    |

### NON -Functional requirements:

| <b>NFR No.</b> | <b>Non-Functional Requirement</b> | <b>Description</b>                                                           |
|----------------|-----------------------------------|------------------------------------------------------------------------------|
| <b>NFR-1</b>   | <b>Usability</b>                  | The system should have a user-friendly and intuitive interface.              |
| <b>NFR-2</b>   | <b>Security</b>                   | Secure login and user data encryption should be implemented.                 |
| <b>NFR-3</b>   | <b>Reliability</b>                | The system should perform consistently under expected loads without failure. |
| <b>NFR-4</b>   | <b>Performance</b>                | Visualizations should load within 3 seconds on average.                      |
| <b>NFR-5</b>   | <b>Availability</b>               | The system should be accessible 24/7 with minimum downtime.                  |
| <b>NFR-6</b>   | <b>Scalability</b>                | Should be scalable to handle large datasets or more branches in the future.  |

### 3.3 Data Flow Diagram



### 3.4 Technology Stack

#### Data layer

| Component             | Description                                                                      |
|-----------------------|----------------------------------------------------------------------------------|
| <b>Dataset</b>        | food_coded.csv – structured CSV dataset capturing dietary and health information |
| <b>Storage Format</b> | Flat file (CSV) loaded locally for analysis                                      |
| <b>Tools Used</b>     | Microsoft Excel (for cleaning and preprocessing if needed)                       |

#### Data processing layer

| Tool/Technology | Purpose                                                            |
|-----------------|--------------------------------------------------------------------|
| <b>Excel</b>    | Initial cleaning or quick field review before importing to Tableau |

#### Visualization & Analytics Layer

| Tool/Technology          | Purpose                                                                  |
|--------------------------|--------------------------------------------------------------------------|
| <b>Tableau</b>           | Main tool for interactive data visualization and dashboard creation      |
| <b>Tableau features</b>  | KPI cards, Bar Charts, Pie Charts, Heatmaps, Highlight Tables, Forecasts |
| <b>Calculated Fields</b> | Used for: Consumption, Food Selection                                    |

#### User Interaction Layer

| Feature | Role |
|---------|------|
|---------|------|



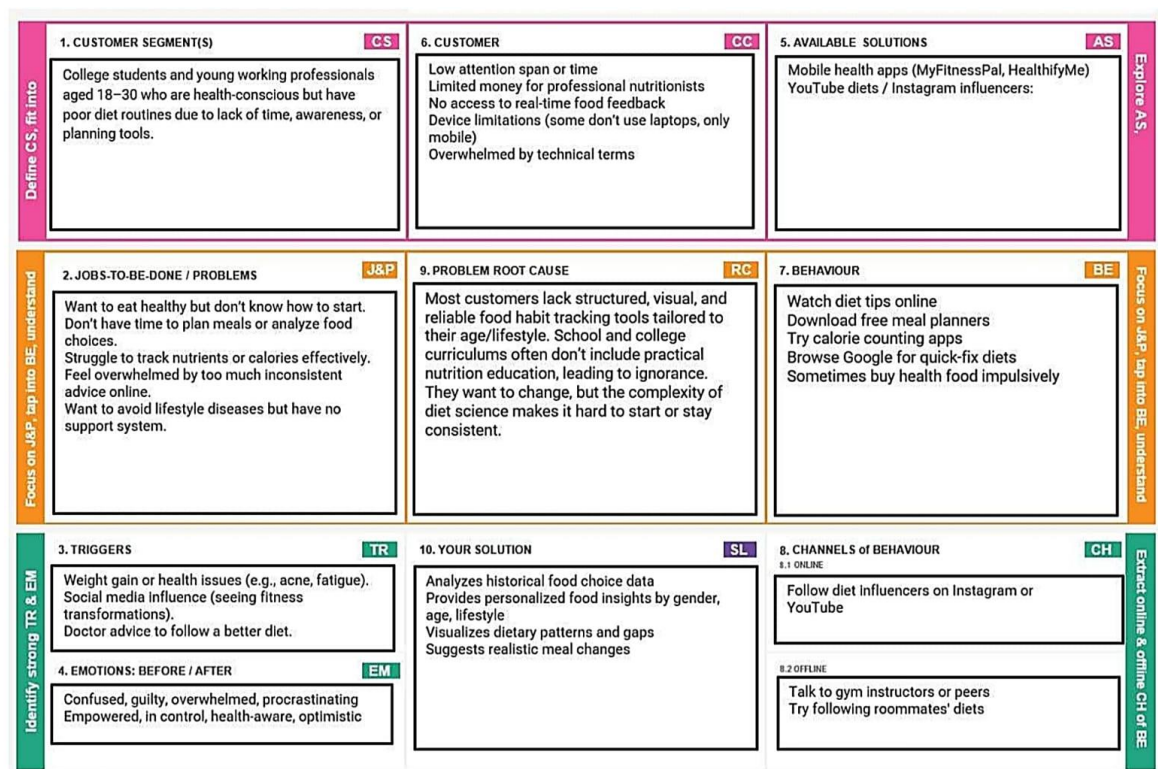
|                               |                                                                           |
|-------------------------------|---------------------------------------------------------------------------|
| <b>Interactive Dashboards</b> | Users can filter data by gender, risk, and exercise levels                |
| <b>Parameter Controls</b>     | Customize target values                                                   |
| <b>Alerts &amp; KPIs</b>      | Instant insight into nutrition deficiencies and trends                    |
| <b>User Roles</b>             | Admin, Nutritionist, Cafeteria Staff – each interacts with filtered views |

## Security & Sharing

| Feature               | Notes                                                 |
|-----------------------|-------------------------------------------------------|
| <b>Tableau Public</b> | Public dashboards (for non-sensitive data)            |
| <b>Tableau Server</b> | Optional upgrade for secured, role-based access       |
| <b>Export Options</b> | PDF reports, public link sharing, dashboard embedding |

## 4.PROJECT DESIGN

### 4.1 Problem Solution Fit



### 4.2 Proposed Solution



| S.No. | Parameter                                       | Description                                                                                                                                  |
|-------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <b>Problem Statement (Problem to be solved)</b> | Clearly describe the problem you aim to solve. Include relevant statistics or pain points to highlight the severity or urgency of the issue. |
| 2     | <b>Idea / Solution Description</b>              | Outline your proposed solution. Focus on how it addresses the problem effectively and efficiently.                                           |
| 3     | <b>Novelty / Uniqueness</b>                     | Explain what makes your solution different from existing ones. Highlight any innovative features, technologies, or processes.                |
| 4     | <b>Social Impact / Customer Satisfaction</b>    | Describe how your solution improves lives, benefits communities, or enhances customer satisfaction.                                          |
| 5     | <b>Business Model (Revenue Model)</b>           | Explain how the solution will generate revenue. Include pricing strategies, partnerships, and target customer segments.                      |
| 6     | <b>Scalability of the Solution</b>              | Discuss how your solution can grow geographically or serve more users. Mention potential challenges and how they will be handled.            |

## 4.3 Solution Architecture

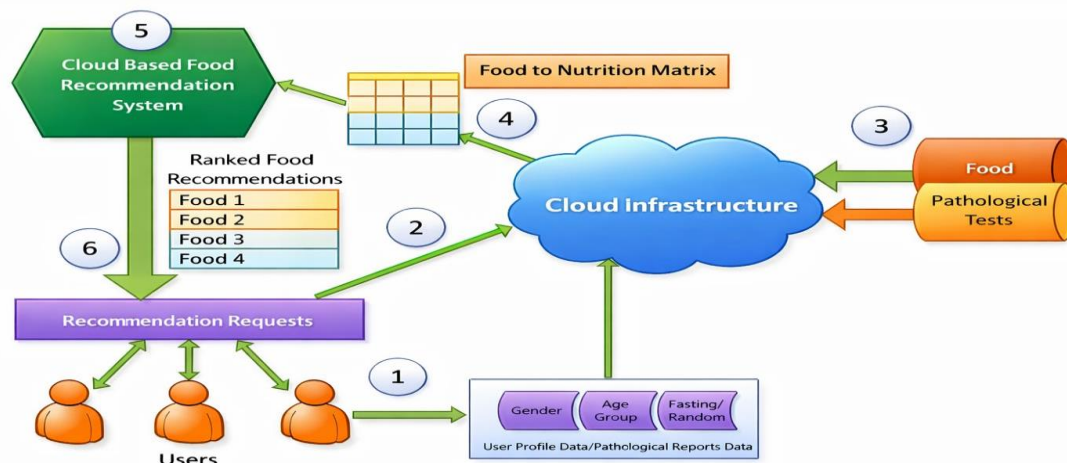


Figure 1: Architecture and data flow of the **Enhanced Dietary Strategies a College Food Choices**

## 5.PROJECT PLANNING AND SCHEDULING

### 5.1 Project Planning

| <b>Sprint</b>   | <b>Functional Requirement (Epic)</b> | <b>User Story Number</b> | <b>User Story / Task</b>                                                                                  | <b>Story Points</b>              | <b>Priority</b>                                        | <b>Team Members</b>                 |
|-----------------|--------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------------------|-------------------------------------|
| <b>Sprint-1</b> | Registration                         | USN-1                    | As a user, I can register for the application by entering my email, password, and confirming my password. | 2                                | High                                                   | 3                                   |
| <b>Sprint-1</b> | Confirmation                         | USN-2                    | As a user, I will receive a confirmation email once I have registered for the application.                | 1                                | High                                                   | 3                                   |
| <b>Sprint-2</b> | Registration                         | USN-3                    | As a user, I can register for the application through Facebook.                                           | 2                                | Low                                                    | 3                                   |
| <b>Sprint-1</b> | Registration                         | USN-4                    | As a user, I can register for the application through Gmail.                                              | 2                                | Medium                                                 | 3                                   |
| <b>Sprint-1</b> | Login                                | USN-5                    | As a user, I can log into the application by entering email & password.                                   | 1                                | High                                                   | 3                                   |
| <b>Sprint</b>   | <b>Total Story Points</b>            | <b>Duration</b>          | <b>Sprint Start Date</b>                                                                                  | <b>Sprint End Date (Planned)</b> | <b>Story Points Completed (as on Planned End Date)</b> | <b>Sprint Release Date (Actual)</b> |
| <b>Sprint-1</b> | 20                                   | 6 Days                   | 01 June 2026                                                                                              | 08 June 2026                     | 20                                                     | 08 June 2026                        |

|                 |    |        |              |              |    |              |
|-----------------|----|--------|--------------|--------------|----|--------------|
| <b>Sprint-2</b> | 20 | 6 Days | 09 June 2026 | 15 June 2026 | 20 | 15 June 2026 |
| <b>Sprint-3</b> | 20 | 6 Days | 16 June 2026 | 22 June 2026 | 20 | 22 June 2026 |
| <b>Sprint-4</b> | 20 | 6 Days | 23 June 2026 | 30 June 2026 | 20 | 30 June 2026 |

## 6.FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

| S.No. | Parameter               | Screenshot / Values                                           |
|-------|-------------------------|---------------------------------------------------------------|
| 1     | Data Rendered           | 125 records                                                   |
| 2     | Data Preprocessing      | Removed null values                                           |
| 3     | Utilization of Filters  | Gender1 Filter                                                |
| 4     | Calculation Fields Used | Usually numeric, or text depending on the formula and context |
| 5     | Dashboard Design        | No. of Visualizations / Graphs – 9                            |
| 6     | Story Design            | No. of Visualizations / Graphs – 1                            |

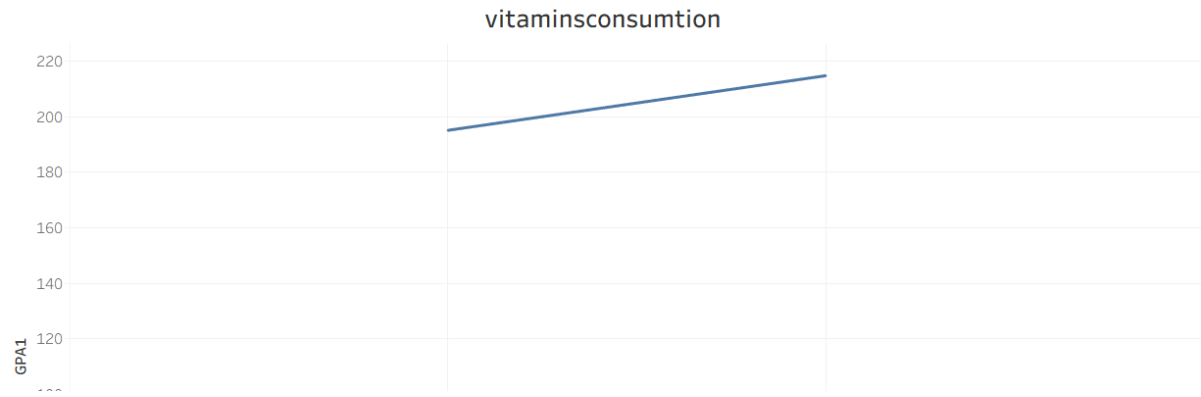
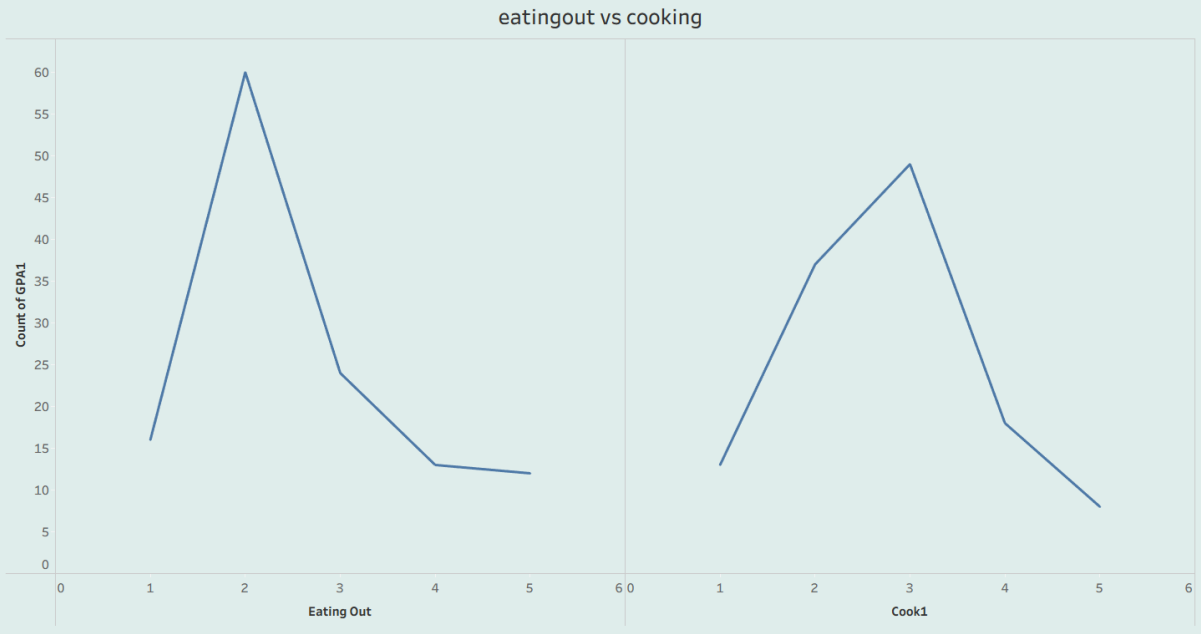
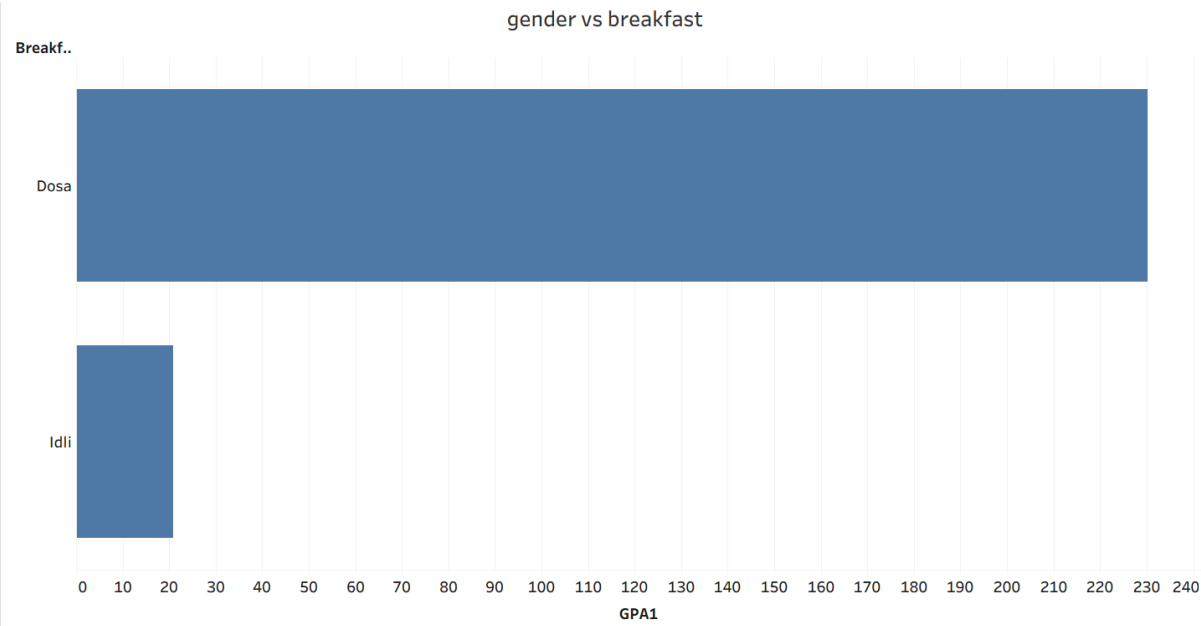
## 7.RESULTS

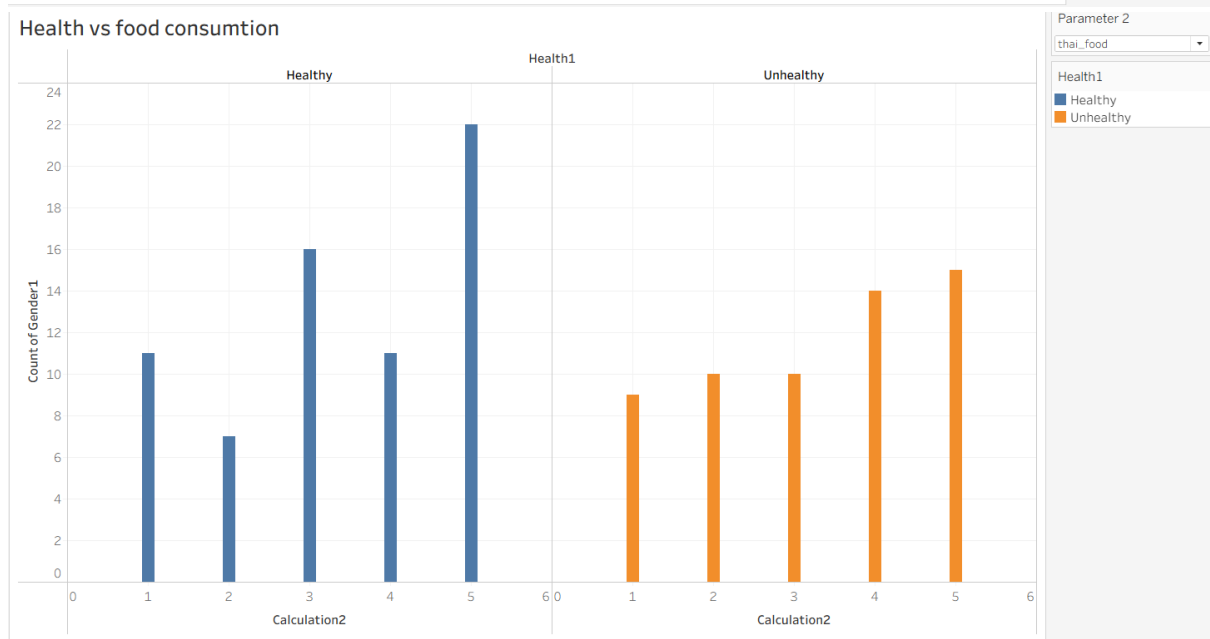
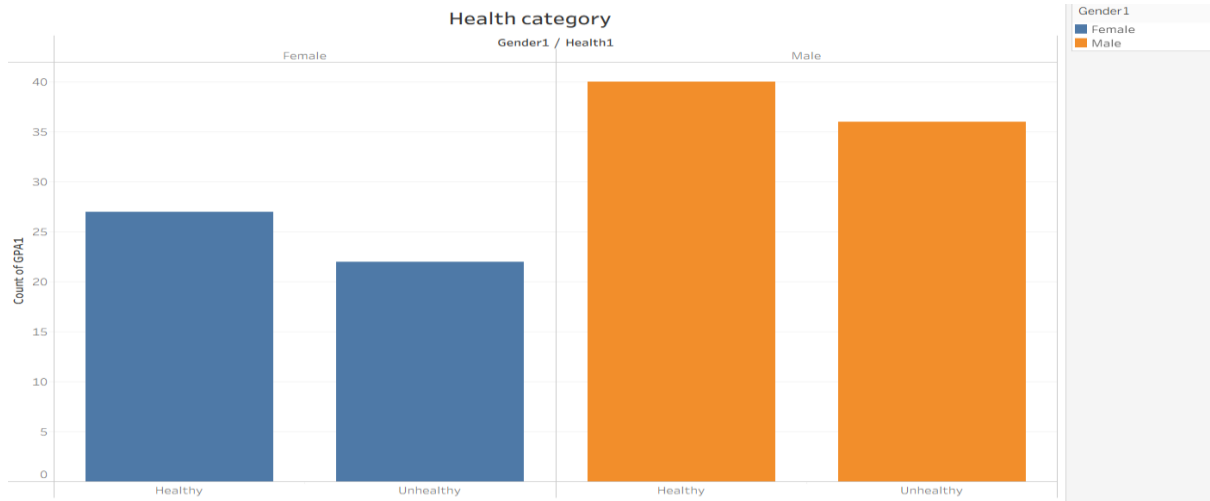
### 7.1 Output Screenshots

KPI Card

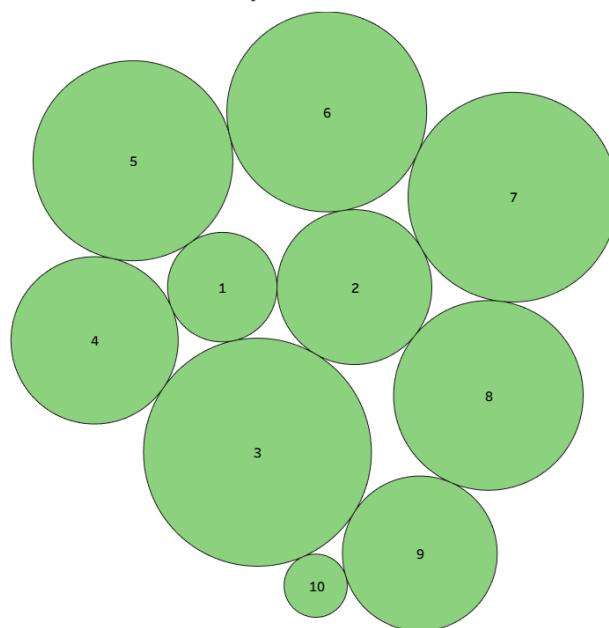
125.0                      3.3                      297.0

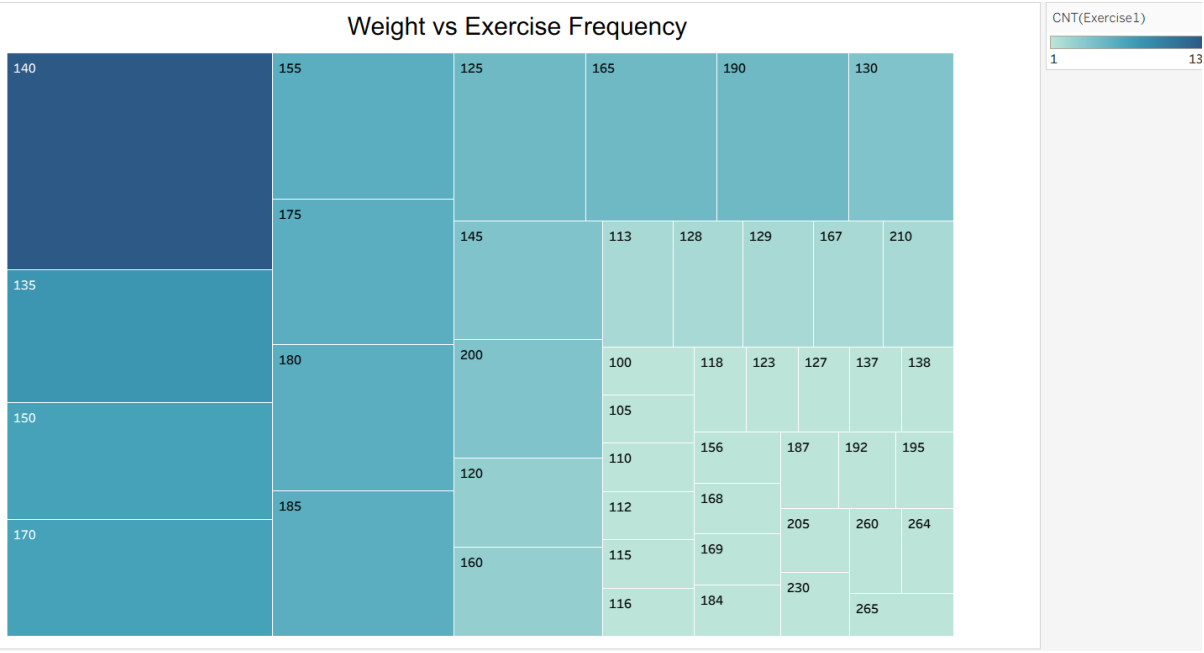
Count of GPA1 Avg. GPA1 Grade Level



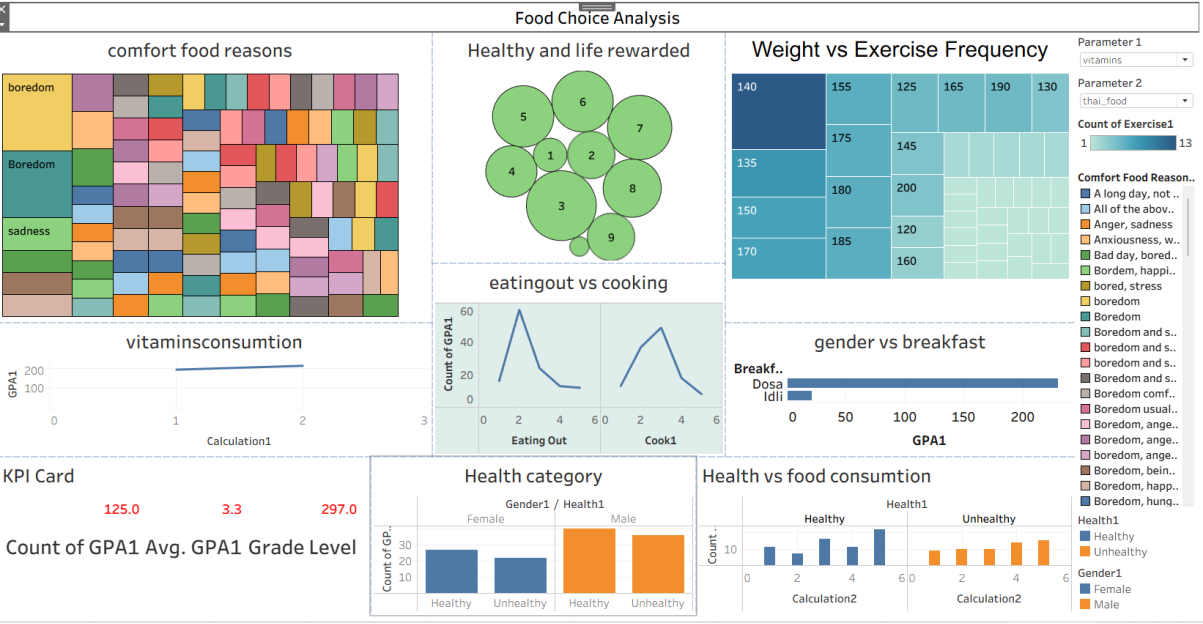


Healthy and life rewarded





Dashboards:







## 8.ADVANTAGES AND DISADVANTAGES

### ADVANTAGES:

- **Real-Time Insights:**

Tableau dashboards allow for up-to-date visualization of student dietary behaviours, helping administrators quickly spot trends and deficiencies.

- **Data-Driven Decision Making:**

Enables evidence-based strategies for improving menu planning, wellness programs, and targeted health interventions.

- **Centralized Data Integration:**

Combines multiple data sources (dining hall menus, nutrition data, student surveys, health records) into one accessible platform.

- **Improved Student Health Outcomes:**

Early identification of dietary deficiencies or poor eating habits can lead to timely support, better physical and mental health, and improved academic performance.

- **Enhanced Collaboration:**

Facilitates cooperation between health services, dining services, and student affairs through shared data insights.

- **Customizable Visual Dashboards:**

Dashboards can be tailored for different audiences—administrators, nutritionists, dining staff, or even students.

- **Cost-Effective Long-Term Planning:**

Helps avoid expensive health interventions later by focusing on prevention through nutrition.

## **DISADVANTAGES:**

- **Data Collection and Quality Issues:**

Accurate and comprehensive data from multiple sources (e.g., dining systems, health records) may be hard to collect, inconsistent, or siloed.

- **Privacy and Ethics Concerns:**

Handling student health and dietary data requires strict compliance with privacy laws (e.g., FERPA, HIPAA) and ethical considerations.

- **Technical and Resource Limitations:**

Building and maintaining Tableau dashboards may require technical expertise, staff training, and ongoing IT support.

- Limited Student Participation:

Without consistent student input (e.g., food choices, survey responses), data may be incomplete or unrepresentative.

- Resistance to Change: 19

Dining services or administrative staff may resist adopting new data systems or changing long-standing menu practices.

- Initial Time and Cost Investment:

Upfront costs for data integration, software, and training may be significant, especially for smaller institutions.

## 9.CONCLUSION

The Comprehensive Analysis and Dietary Strategies with Tableau project represents a forward-thinking approach to improving student health and wellness through data-driven insights. By leveraging Tableau's powerful visualization capabilities, the project addresses key challenges such as fragmented data, lack of real-time analysis, and reactive health strategies. With integrated dashboards and nutritional analysis tools, college administrators, health professionals, and dining services can collaboratively identify dietary trends, detect nutritional gaps, and implement more effective, targeted interventions.

This initiative not only enhances operational efficiency but also places student well-being at the centre of campus life—fostering a healthier, more informed, and proactive campus community.

## 10.FUTURE SCOPE

1. Predictive Health Analytics: o Integrate machine learning tools with Tableau to forecast potential health issues related to diet (e.g., vitamin deficiencies, obesity risk).
2. Mobile App Integration: o Develop or integrate with mobile platforms to allow students to track their meals, receive personalized nutrition tips, and give feedback.
3. Personalized Nutrition Plans: o Use student dietary data to offer tailored meal recommendations based on individual health needs, allergies, or fitness goals.

4. Expansion to Mental Health Insights: o Correlate dietary trends with mental health patterns to support holistic student wellness initiatives.
5. Collaboration with Academic Departments: o Partner with nutrition, public health, or data science departments for student- led research, internships, or capstone projects.
6. Benchmarking Across Institutions: o Share anonymized insights across universities to compare trends, share best practices, and improve nationwide student health strategies. 20
7. Sustainability & Food Waste Monitoring: Extend dashboards to analyse food waste and sustainability metrics, aligning dietary planning with environmental goals

## 11.APPENDIX

Dataset link:<https://www.kaggle.com/datasets/borapajo/food-choices>

GitHub repository link:[https://github.com/msilas1810-cyber/Comprehensive-Analysis-and-Dietary-Strategies-with-Tableau-\\_A-College-Food-Choices-Case-Study.git](https://github.com/msilas1810-cyber/Comprehensive-Analysis-and-Dietary-Strategies-with-Tableau-_A-College-Food-Choices-Case-Study.git)

Project demo

link:[https://drive.google.com/file/d/1ZjhZ\\_Jj9jdAHoZden1SWvJT6ZLI52Z4s/view?usp=drive\\_link](https://drive.google.com/file/d/1ZjhZ_Jj9jdAHoZden1SWvJT6ZLI52Z4s/view?usp=drive_link)